

Assignment 1: Matlab Basics

1. The volume V of liquid in a hollow horizontal cylinder of radius r and length L is related to the depth h of the liquid by

$$V = \left[r^2 \cos^{-1} \left(\frac{r-h}{r} \right) - (r-h) \sqrt{2rh - h^2} \right] L$$

Develop an M-file to create a plot of volume (V) versus depth (h). Note that r and L are specified as constants and h cannot be greater than $2r$. Test the program for $r = 2$ m and $L = 5$ m.

2. The equation representing a falling bungee jumper can be derived via Newton's second law as:

$$\frac{dv}{dt} = g - \frac{c}{m} v^2$$



Where v is the downward velocity, t is the time, g is the gravitational acceleration, c is the air drag coefficient, and m is the mass of the jumper. Write a MATLAB code to solve this modified problem numerically using the forward Euler method for $t = 0$ to 12 seconds and plot v vs t . Assume the following values: $g = 9.8$ m/s², $c = 0.25$ kg/m, and $m = 68.1$ kg. Assume an initial condition of $v = 0$ at $t = 0$.

HINT: You can refer to the example MATLAB script `parachute.m` (on moodle) and the lecture notes for help for this problem and the next two problems.

3. The following equation is a simplified form of bioremediation kinetics that may occur in a bioreactor.

$$\frac{dc}{dt} = -\mu X \left(\frac{c}{K + c} \right)$$

where

c = concentration of contaminant or substrate (mg/L)

K = Monod half saturation constant = 2 mg/L

$X = \text{biomass concentration} = 0.5 \text{ mg/L}$

$\mu = \text{maximum specific substrate utilization rate} = 4 \text{ day}^{-1}$

Assume an initial condition of $t = 0$, $c = c_0 = 2 \text{ mg/L}$. Note that X is usually not constant but for simplicity we will assume so.

Write a MATLAB code to solve the above equation analytically (using 'dsolve') and numerically (using forward Euler) for $c(t)$ and compare the results in a plot. Solve the equation up to a time of 12 days.

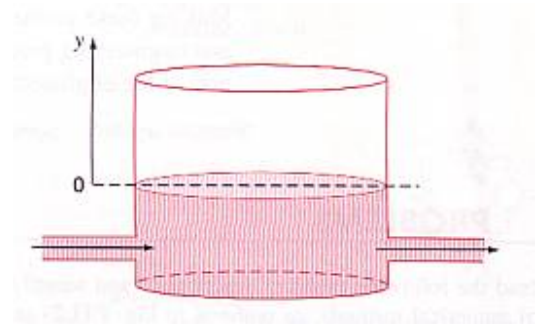
4. A storage tank contains liquid at depth y , where $y = 0$ in the beginning. Liquid is withdrawn at a constant flow rate Q to meet demands. The contents are resupplied at a sinusoidal rate $3Q\sin^2 t$. Continuity equation can be written for this system as:

$$\left(\begin{array}{c} \text{rate of change} \\ \text{of volume} \end{array} \right) = (\text{inflow rate}) - (\text{outflow rate})$$

$$\frac{d(Ay)}{dt} = 3Q\sin^2 t - Q$$

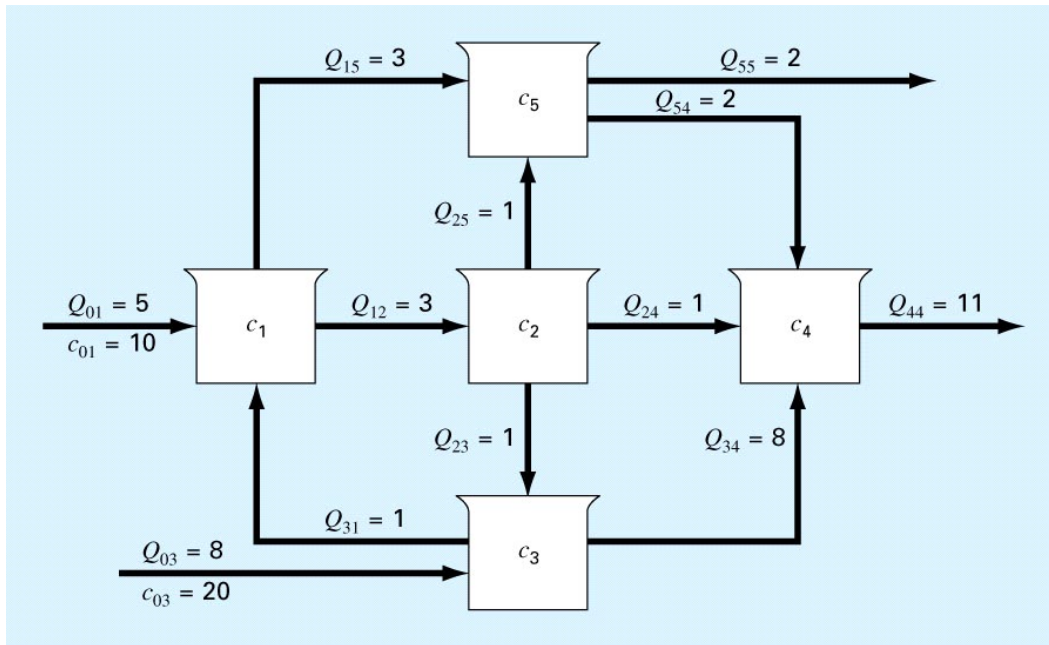
or, if the surface area, A , is constant,

$$\frac{dy}{dt} = 3\frac{Q}{A}\sin^2 t - \frac{Q}{A}$$

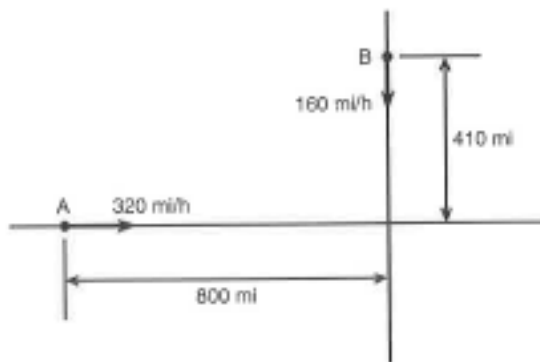


Again use the forward Euler method to solve this problem numerically for the height y from $t = 0$ to 5 days with a step size of 0.5 day. The parameter values are $A = 1200 \text{ m}^2$ and $Q = 400 \text{ m}^3/\text{day}$.

5. Write a MATLAB code to estimate the steady state tank concentrations (c_1, c_2, c_3, c_4, c_5) in the reactor configuration. You can use the following mass balance relationship (input mass = output mass) assuming that the reactors behave as completely mixed reactors: $\sum Q_{i,in} C_{i,in} = \sum Q_{i,out} C_i$. You will get 5 linear equations and 5 unknowns. Use the MATLAB's backslash ' $\backslash$ ' operator to solve the resulting matrix system.



6. Aircraft A is flying east at 320 mi/hr, while aircraft B is flying south at 160 mi/hr. At 1:00 pm, the aircraft are located as shown in the figure below.



- Obtain the expression for the distance D between the aircraft as a function of time. Plot D versus time until after D reaches its minimum value.
- Use this plot to approximately determine the time when the aircraft are first within 30 mi of each other (T_{30}). Check your graphical solution (T_{30}) with the solution obtained by MATLAB's `fzero` function.

7. (i) Read the following excel file '[energy_generation_nc.xlsx](#)' using `readcell` and plot a pie chart of energy utilization by source. The third column in the excel file includes the labels for the pie chart.

(ii) Read the following excel file '[energy_generation_all_states.xlsx](#)' using `readcell` and plot pie chart of energy utilization by source for a given state. Use 'input' to input the state from keyboard and use '`strcmp`' function to extract the data pertaining to the given state.